



MULUNGUSHI UNIVERSITY

Pursuing the frontiers of knowledge

School of Engineering and Technology

ICT312 – INTRODUCTION TO COMPILERS

2024/2025 Academic Year - Semester II

June 2025 Examination

Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES

1. There are two sections in this question paper: **Section A** and **Section B**.
2. You are required to attempt **ALL** questions in **Section A**, and any **three (3)** questions from **Section B**.
3. Read all questions carefully before attempting to answer
4. Begin every question on a new page of the answer booklet.
5. For questions requiring calculations, show your working clearly and neatly.
6. All diagrams should be well labeled.
7. Indicate your Student Number and NRC number on the answer booklet provided.
8. Cell phones and programmable calculators are not allowed in the examination room.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION A: Compulsory Questions (40 marks)
Attempt ALL questions in this section.

Question 1 [40 Marks]

- Describe a compiler in a single sentence, capturing its essence, purpose, and role within software development? (5 marks)
- List **three** key differences between a compiler and an interpreter. (6 Marks)
- The following diagram shows phases of a compiler. Label the missing phases (A, B, C, D).
 $\text{Source Code} \rightarrow [A] \rightarrow [\text{Syntax analysis}] \rightarrow [B] \rightarrow [\text{Intermediate Code Generation}] \rightarrow [C] \rightarrow [\text{Code Generation}] \rightarrow [\text{Symbol Table Management/Assembly}] \rightarrow [D]$
 - Identify phases A, B, C, D. (8 Marks)
 - Which phase checks for **type compatibility**? (3 Marks)
 - Which phase **removes redundant code**? (3 Marks)
 - What is the output of **Lexical Analysis** called? (3 Marks)
 - Give **two examples** of errors detected in **Syntax Analysis**. (4 Marks)
- Consider the following grammar below.
 $A \rightarrow ABd / Aa / a$
 $B \rightarrow Be / b$
 - Define a left recursive grammar (3 marks)
 - Eliminate left recursion in the grammar above (5 marks)

SECTION B: Attempt ANY three (3) questions (60 marks)

Question 2 [20 Marks]

- Consider the following arithmetic expression in a high-level programming language:
 $\text{result} = (\text{value1} + \text{value2}) * 3.142 - \text{offset};$
 - Identify and list all the lexemes in the expression. (3 marks)
 - For each of the following phases of compilation, describe the output produced when processing this expression:
 - Lexical Analysis (3 marks)
 - Syntax Analysis (3 marks)
 - Semantic Analysis (3 marks)
 - Intermediate Code Generation (2 marks)
- Modern compilers separate **lexical analysis** (scanning) from **parsing** (syntax analysis).
 - Discuss** three key advantages of this separation, supporting your answer with brief explanations. (6 marks)

Question 3 [20 Marks]

- Suppose a compiler C1 (written in language L1) compiles source code in L2. If you want to create a compiler C2 for L2 written in L2, explain the bootstrapping steps. (5 Marks)
- Write a CFG for strings over {a, b} where every a is immediately followed by at least one b. (5 marks)

- bababab

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G^*
 $G \times G$
 $G \times G \times G$

$$A \rightarrow c/\varepsilon$$

$\begin{matrix} S & A & B \\ \downarrow & \downarrow & \downarrow \\ Z & E & E \\ N & P & P \\ A & B & C \end{matrix}$

- [illegible]

$\begin{matrix} S & A & B \\ \downarrow & \downarrow & \downarrow \\ Z & E & E \\ N & P & P \\ A & B & C \end{matrix}$

- $\begin{matrix} S & A & B \\ \downarrow & \downarrow & \downarrow \\ Z & E & E \\ N & P & P \\ A & B & C \end{matrix}$

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